Outcome Composition Can Invert Do-PFN's Uplift Ranking: A Criteo Case Study

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ABSTRACT

Causal foundation models such as Do-PFN and CausalPFN estimate treatment effects in context: they read a sample of labeled rows and score new users without training. For uplift targeting on large logs, the rows that fill the context are a design choice that causal PFN evaluations seldom vary. On the Criteo uplift benchmark, where about 5% of users visit, the released Do-PFN checkpoint orders users worse than random when its 1,024-row context follows the natural outcome rate: normalized Qini is −0.077 in cross-validation and −0.046 on a 2.09M-row holdout. Balancing outcomes within each treatment arm reverses the sign without retraining (+0.056 and +0.048). The inversion lies in how Do-PFN reads the context: CausalPFN, a TabPFN T-learner and classical learners fitted on the same rows do not invert. Across outcome shares, the sign mostly tracks how Do-PFN's scores align with baseline visit risk (one exception, at a 35% share); Criteo's uplift is concentrated among the highest-risk users. A 20% share tuned on cross-validation beats balancing on a holdout subsample (+0.025 over 10 contexts; two-level bootstrap interval [0.005, 0.043]) but is not better on a second, noisier dataset. Binary outcomes lie outside the checkpoint's documented regime, so we report a diagnosis and a cheap repair, not a verdict on the method.

CCS CONCEPTS

• **Computing methodologies** → **Causal reasoning and diagnostics**; • **Information systems** → *Online advertising*.

KEYWORDS

uplift modeling, causal foundation models, in-context learning, prior-data fitted networks, class imbalance

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1 INTRODUCTION

Uplift models rank users by how much a treatment, such as an ad impression or a coupon, changes their chance of acting, so that a campaign reaches the users it persuades [6]. The usual practice is to fit a meta-learner for each campaign [10]. Causal prior-data fitted networks (PFNs) offer another workflow. Do-PFN [15] and CausalPFN [2] are pretrained on simulated causal data. Like the TabPFN classifiers they build on [7, 8], they read a context of labeled rows (x, t, y) and return effect estimates for new users in one forward pass. On web-scale logs the context is a tiny sample, here 1,024 of almost 12M development rows, so the sampler that fills it is part of the method. For predictive PFNs, choosing and tuning the context is an active topic [5, 16].

For rare binary outcomes the sampler matters. Case-control sampling, which over-represents positives, shifts the intercept of a logistic model but leaves its slopes intact [9, 13]. In uplift modeling, undersampling the majority class helps some learners and hurts others [12]. For PFN classifiers, downsampling the majority class works about as well as the best correction [11]. We ask the same question for causal PFNs and get a sharper answer than "balancing helps".

On the Criteo uplift benchmark [4], 4.7% of users visit. When each arm of the context keeps its natural visit rate (about 4% of context rows), the released Do-PFN checkpoint orders users worse than random in every cross-validation fold and on a frozen 2.09M-row holdout. We fill each treatment arm of a context of the same size with equal numbers of visitors and non-visitors (within-arm outcome balancing), and the sign turns positive without retraining (Fig. 1a). We ask whether the outcome composition of a fixed-size context, rather than its size, sets the sign and whether the repair holds on a holdout (RQ1); whether the inversion belongs to the context rows or to the model that reads them (RQ2); whether the sign tracks how Do-PFN's scores align with baseline risk (RQ3); and whether the share can be tuned into a reliable, competitive ranker (RQ4).

We contribute a diagnosis and a cheap repair for a public causal foundation model. The sampler itself is standard; what is new is that its choice flips the sign of a causal PFN's ranking. Do-PFN documents continuous outcomes as its regime, so our results describe the released checkpoint on a binary benchmark and do not refute the method.

2 SETUP

Data. Criteo uplift v2.1 [4] pools randomized ad-exposure trials: 13.98M users, 85% treated, anonymized features and a binary visit outcome. The cleaned copy we use lacks two of the 12 features (f7, f10). A 15% holdout (2,093,918 rows), split by groups of identical feature vectors, was set aside before modeling. On it, treated users visit at 4.87% and controls at 3.86%, an average effect of +1.01 percentage points (pp). The other rows form five cross-validation (CV) folds with the same grouping; each fold is evaluated on 8,000 of its rows. Of the five uplift datasets in our CV grid, Do-PFN's $\hat{Q}$ under T contexts is clearly negative only on Criteo. It is within noise on LZD (3.5% positive; −0.016) and Hillstrom (15%; −0.005), and slightly positive on RetailHero (62%) and Orange (97%). We

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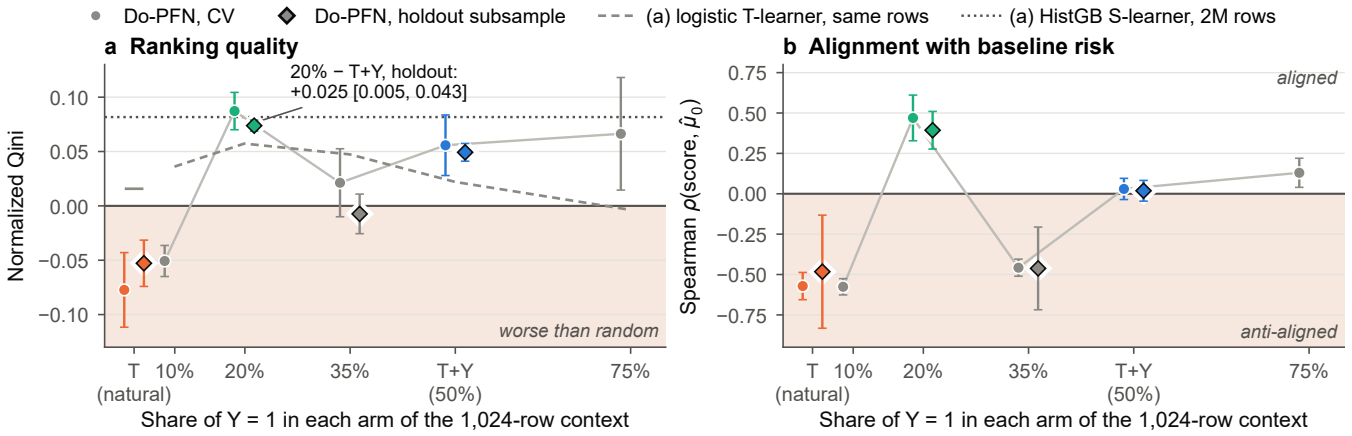


Figure 1: Do-PFN against the share of visitors (Y=1) in each treatment arm of a 1,024-row Criteo context. Circles: CV fold means with 95% fold-level t-intervals. Diamonds: the 200k-row holdout subsample, t-intervals over contexts (3 for T and 35%, 10 for 20% and T+Y); both are offset for legibility. (a) Ranking quality; the dashed line shows CV means (a short dash marks T), the dotted one a holdout-subsample score, the annotation a two-level bootstrap interval. (b) Rank correlation of Do-PFN's scores with estimated baseline risk $\hat{\mu}_0$. $\hat{Q}$ is at or below zero wherever $\rho < 0$, except at 35% in CV.

therefore study Criteo and use LZD, the other dataset with rare positives (the randomized Lazada split released with DESCN [17]; 181,669 rows), as a transfer check.

Models. *Do-PFN* [15], released checkpoint. The binary outcome is its regression target (z-scored, as the official interface does), and a user's score is $\hat{\tau}(x) = \hat{\mathrm{E}}[Y \mid do(T{=}1), x] - \hat{\mathrm{E}}[Y \mid do(T{=}0), x]$. The released *CausalPFN* [2] checkpoint, and a *TabPFN T-learner* [8] with one TabPFN classifier per arm. As controls fitted on exactly the same context rows: a *logistic T-learner* and a histogram gradient-boosting (*HistGB*) *S-learner*. As a data-rich reference for the holdout: a HistGB S-learner fitted on 2M development rows.

Context designs. Each context holds K rows from the training fold; for the holdout they come from development rows. R samples uniformly. T takes half treated and half control rows, with outcomes at their natural rate of about 4%. $T{+}Y$ makes half the rows in each arm visitors. *Share* s makes a fraction s of each arm visitors, so T+Y is s=50%. Equal-share designs show the model a zero average effect; the 10% design does so and still inverts, so the repair is not the zeroed average itself. We use K=1,024 and check K from 512 to 4,096.

Metrics. The normalized Qini $\hat{Q}$ [14] is the area between a model's Qini curve and the random-targeting line, divided by the same area for the optimal ordering: 0 is random, negative is worse than random. AUUC, the area under the uplift curve, is normalized the same way. Uplift@10% is the treated minus control visit rate among the 10% top-scored users, in pp. All three are computed with scikit-uplift. We call a ranking *inverted* when $\hat{Q}$ is negative with an interval excluding zero. In CV we average three context seeds per fold and report a t-interval over the five folds (4 degrees of freedom, ignoring the overlap of training folds, so descriptive). On the holdout we report t-intervals over context seeds, paired by seed; a cluster bootstrap over identical feature vectors adds row-level noise. To relate scores to baseline risk, we fit HistGB on the control arm of rows disjoint from the scored ones, which gives $\hat{\mu}_0(x)$: held-out rows for the CV analyses and 2M development rows for the holdout analysis. We report the Spearman correlation $\rho(\hat{\tau}, \hat{\mu}_0)$.

Holdout use. The comparison of T and T+Y on the full holdout was run once. Holdout labels also fit the risk model behind the CV correlations that motivated our holdout hypotheses, so the holdout is held out from tuning, not untouched. The share follow-up (RQ3–4) uses a fixed random 200k-row subsample of the holdout, with hypotheses written into an internal protocol shortly before the runs, without a hash. Its first check of 20% against T+Y used three contexts and gave +0.024, with a t-interval over contexts of $[-0.003, 0.052]$. We then added seven contexts, so the 10-context comparison is a follow-up, not a replication.

3 RQ1: COMPOSITION SETS THE SIGN

Under R or T contexts, Do-PFN's normalized Qini is negative at every context size (Fig. 2a): −0.072 to −0.085 for K from 512 to 4,096. Under T at K=1,024 it is −0.077, negative in all five folds and in 14 of 15 runs. Outcome balancing gives +0.037 to +0.056 at the same sizes. At K=1,024 the gain over T is +0.133 [0.102, 0.164] (Fig. 2b). On LZD the signs match (T −0.016, T+Y +0.016, gain +0.032 [−0.027, 0.090]), but none is resolved.

The repair holds out of sample (Table 1). On all 2.09M holdout rows, $\hat{Q}$ rises from −0.046 under T to +0.048 under T+Y: a gain of +0.094 [0.037, 0.150] over three contexts per design. The gain in uplift@10%, +1.8 pp, is not resolved with three contexts. The inversion lies in the overall order, not at the very top. Under T the 10% top-scored users still show above-average uplift (1.3–2.3 pp against 1.01 pp), but the 10% bottom-scored show more (2.3–2.9 pp); under T+Y these are 2.7–4.0 and 0.3–0.7 pp.

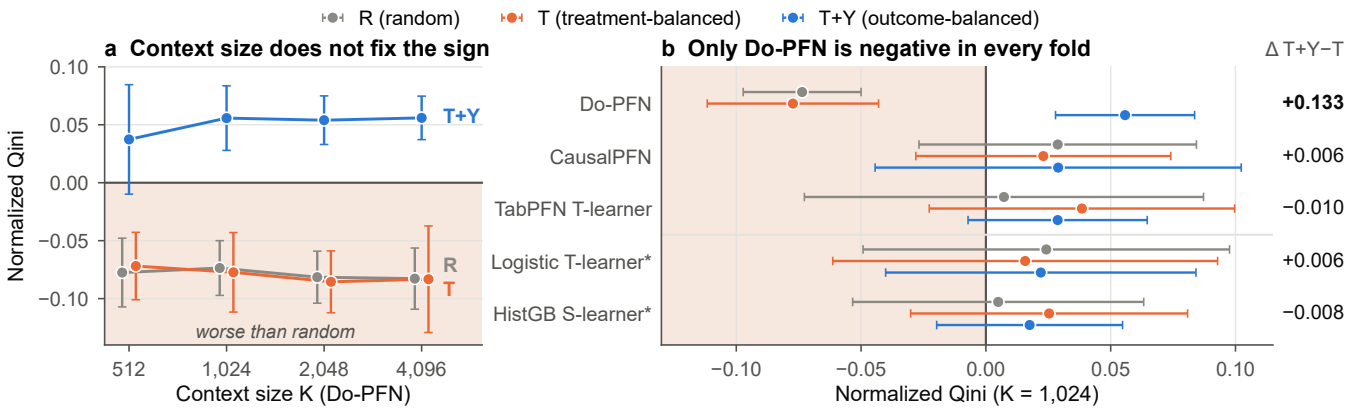


Figure 2: Criteo CV: fold means with 95% fold-level t-intervals; shading marks rankings worse than random. (a) Do-PFN against context size (R and T offset slightly). (b) The same K=1,024 contexts read by three foundation models and by two learners fitted on those rows (*); under R and T, no model other than Do-PFN is negative in more than two of five folds. Right column: gain from balancing, T+Y minus T; only Do-PFN's is resolved.

Table 1: Do-PFN on the frozen Criteo holdout, K=1,024. Rows show means over n contexts; contrasts show 95% t-intervals over contexts, paired by context seed.

Design	n	$\hat{Q}$	AUUC	Uplift@10% (pp)
Full holdout, 2,093,918 rows				
T	3	−0.046	−0.018	1.76
T+Y	3	0.048	0.019	3.56
T+Y − T		**+0.094**	+0.037	+1.80
		[0.037, 0.150]	[0.015, 0.059]	[−1.28, 4.88]
Fixed 200k-row subsample				
T	3	−0.053	−0.021	1.43
35%	3	−0.007	−0.003	2.12
T+Y	10	0.049	0.019	3.50
20%	10	0.074	0.029	4.67
20% − T+Y		**+0.025**	+0.010	+1.17
		[0.015, 0.034]	[0.006, 0.014]	[0.32, 2.02]
Reference on the same 200k rows, fitted on 2M development rows				
HistGB S-learner	–	0.082	0.032	5.36

4 RQ2: THE MODEL, NOT THE ROWS

A context without information about who responds would give $\hat{Q}$ near zero, not below it. Learners fitted on the same rows do give near zero: the logistic T-learner and the HistGB S-learner score +0.005 to +0.025, every interval includes zero, and none is negative on average (Fig. 2b). CausalPFN and the TabPFN T-learner read the same contexts without a detectable inversion either. Do-PFN's resolved negative $\hat{Q}$ therefore needs a systematic source in how it reads the rows. Balancing shows no detectable change for the other two PFNs (+0.006 and −0.010, intervals about ±0.10), while Do-PFN's gain exceeds theirs by +0.127 [0.044, 0.210] and +0.143 [0.062, 0.223] (fold-level interaction). With 8,000 evaluation rows per fold, uplift metrics are noisy [3], so Fig. 2b does not rank the controls.

5 RQ3: THREE ALIGNMENT STATES

With K fixed, Do-PFN's response to the visitor share of each arm is not monotone (Fig. 1). In CV, $\hat{Q}$ is −0.051 at 10%, peaks at +0.087 at 20%, dips to +0.021 at 35%, and recovers to +0.056 and +0.066 at 50% and 75%. The dip is reproduced on the holdout: in all three contexts tried, 35% scores 0.082 [0.058, 0.107] below the three first-check 20% contexts. The logistic T-learner on the same rows also peaks at 20% and gains as much over T+Y there as Do-PFN does (+0.035, interval ±0.11, against +0.031 [0.006, 0.057]; HistGB S-learner +0.003), so the 20% gain may belong to the rows. RQ2's model-specific result concerns T against T+Y.

The sign tracks one property of the scores: their rank correlation with estimated baseline risk $\hat{\mu}_0$ (Fig. 1b). We see three states. *Anti-aligned* (R, T, 10%, 35%): $\rho < 0$ in all 15 CV runs of each design and in every holdout context for T and 35% ($\rho \approx -0.47$). *Aligned* (20%, and weakly 75%): $\rho > 0$ in all 15 CV runs and all 10 holdout contexts at 20% (mean +0.47 in CV, +0.39 on the holdout); +0.13 at 75%. *Mixed* (T+Y): a global $\rho \approx 0$ hides a non-monotone profile (Fig. 3b). Deciles 9–10 sit at the 61st–63rd score percentile, but so does decile 1, and deciles 6–7 sit lowest (31st–33rd).

Figure 3 shows how this alignment relates to the sign on Criteo. Because $\hat{\mu}_0$ is fitted on rows disjoint from those it stratifies, the deciles avoid the bias of endogenous stratification [1]. Observed uplift is concentrated among high-risk users: 2.4 and 4.3 pp in the top two deciles of $\hat{\mu}_0$, and close to zero below decile 8. Under T, Do-PFN places the top decile at a mean score percentile of 24, among the users it would target last; T+Y moves it to 63, and 20% to 87.

This is an association, not a mechanism. At 35% in CV, ρ is negative while $\hat{Q}$ is positive. On LZD, 10% and 20% are aligned (ρ = +0.19, +0.30) yet $\hat{Q}$ is slightly negative, though within noise.

6 RQ4: TUNING THE SHARE, AND ITS LIMITS

We selected the 20% share on CV. On the holdout subsample it beats T+Y in all 10 seed-matched pairs of contexts, with $\hat{Q}$ 0.074

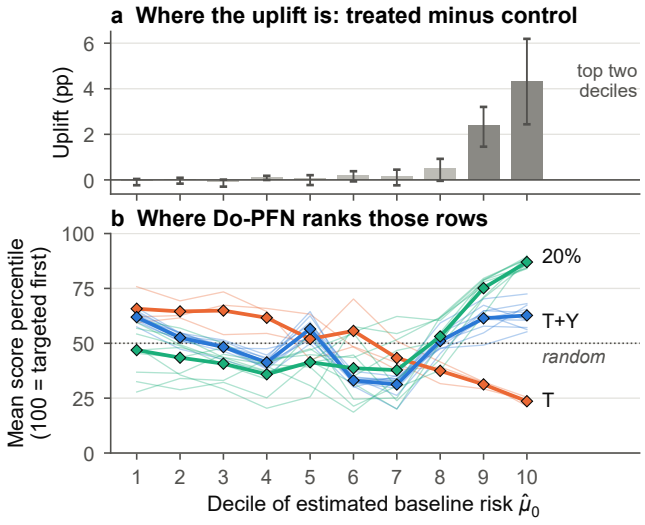


Figure 3: Anatomy of the inversion on the 200k-row holdout subsample. (a) Observed uplift by decile of estimated baseline risk $\hat{\mu}_0$, with Newcombe 95% intervals; $\hat{\mu}_0$ is heavily tied below decile 5, and ties are split by row order. (b) Mean percentile of Do-PFN's score within each decile; thin lines are single contexts (3 for T, 10 for T+Y and 20%).

against 0.049 (gain +0.025 [0.015, 0.034]) and uplift@10% higher by 1.2 pp [0.3, 2.0] (Table 1). Resampling contexts and row clusters together (a two-level bootstrap) keeps the $\hat{Q}$ gain positive ([0.005, 0.043]) but not the uplift@10% gain ([−0.1, 2.4] pp). One limit argues against adopting 20%; two apply more widely.

The tuned share does not transfer. On LZD, 20% is not better than T+Y (−0.026 [−0.075, 0.022] in CV).

A data-rich learner stays slightly ahead. At 20%, Do-PFN's $\hat{Q}$ is 0.008 below that of a HistGB S-learner fitted on 2M development rows; the gap is resolved over contexts [0.003, 0.013] but not once rows are resampled ([−0.006, 0.022]). T+Y (0.049) trails further.

Scores rank but do not estimate. Under T the mean predicted effect is −5 to −10 pp, driven by heavy tails; at 20% it is +6.8 pp with a median near zero, against an observed +1.0 pp.

7 DISCUSSION AND LIMITATIONS

For Do-PFN-style models on data like Criteo, our results suggest three rules. (1) Before trusting a causal PFN's ranking, check its sign on held-out randomized data and try an outcome-balanced context, which costs nothing at inference time. (2) Report the sampler and evaluate several contexts per design: on the full Criteo holdout (T contexts, K=4,096), the context alone moves CausalPFN and the TabPFN T-learner by a standard deviation of 0.04–0.05 in $\hat{Q}$. (3) Use the scores to rank, not as effect estimates.

The evidence has clear bounds: one dataset with a reliable signal, one released checkpoint, contexts of at most 4,096 rows, a holdout that was not untouched, a follow-up added after a first check, two missing features, and a binary outcome outside Do-PFN's documented regime. Why the alignment with risk flips remains open; per-arm predictions, other checkpoints and retrieved contexts [16] are natural next steps.

ETHICS STATEMENT

We use public, anonymized data from randomized trials and collect no new data. Uplift models decide who receives ads or incentives. An inverted ranking wastes budget by placing users who respond more below users who respond less, so a model's ranking direction should be validated on randomized holdout data before deployment. Generative AI tools assisted with analysis code, figure code and drafting of the text; the authors checked all results and the final text.

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